

Lecture Notes of Information Geometry

PhD Course

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Chapter 1

Affine Geometry

1.1 Probability Simplex

A set where each point represents the probability distribution among the finite number of mutually exclusive events.

It is the set of all probability measures on a finite sample space $\mathcal{X}$ is identified with the set $\mathcal{P} = \mathcal{P}(\mathcal{X})$ of all *probability functions* q ,

$$\mathcal{P}(\mathcal{X}) = \left\{ q: \mathcal{X} \ni x \mapsto q(x) \in \mathcal{R}_+^{\mathcal{X}} \mid \sum_{x \in \mathcal{X}} q(x) = 1 \right\} \quad (1.1)$$

In the equation 1.1 every element of $q \in \mathcal{P}$ is non-negative. The set $\mathcal{P}$ is called *standard simplex* of $\mathcal{R}^{\mathcal{X}}$. It is a closed convex set of the vector space of all real functions on $\mathcal{X}$, denoted $\mathcal{R}^{\mathcal{X}}$. A *polytope* is the convex hull of a finite number of points in a vector space. The generating points of the standard simplex are the delta probability functions δ_a , $a \in X$, $\delta_a(x) = (x = a)$. A *simplex* is the polytope generated by *affinely independent* points. All simplexes with the same number of vertexes are transformed into one another by a suitable affine function. A general reference for convex geometry is [6]. The probability simplex in $\mathcal{R}^3$ can be seen in figure 1.1.

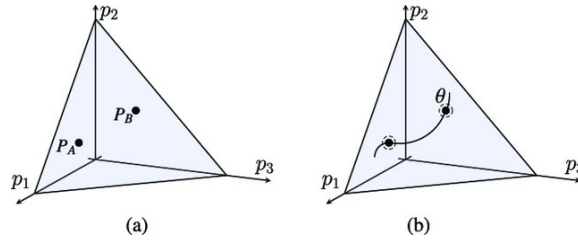


Figure 1.1:

- (a) The probability simplex where the measurement has three outcomes with probabilities p_1, p_2 , and p_3 . Every probability distribution is represented by a point in the simplex (P_A and P_B). (b) h parametrizes a path through the probability simplex. [11]

1.1.1 Affine Space

The *affine space* [6]

$$\mathcal{A}(\mathcal{X}) = \left\{ v \in \mathcal{R}^{\mathcal{X}} \mid \sum_{x \in \mathcal{X}} v(x) = 1 \right\}. \quad (1.2)$$

It is sometimes convenient to ignore the positivity constraint for probability functions and let the "probabilities" be free to assume negative values.

1.1.2 Tangent Space

The *tangent space* is $TP(\mathcal{X}) = \{v \in \mathcal{R}^{\mathcal{X}} \mid \sum_{x \in \mathcal{X}} v(x) = 0\}$. The tangent space is the collection of all possible *velocities* of curves, that is, one-dimensional models, in the simplex or the affine space. If the curve

$$I \ni t \mapsto q(t) \in \mathcal{A}(\mathcal{X}), \quad (1.3)$$

where I is an open real interval, is differentiable at $\bar{t}$ as a function with values in $\mathcal{R}^{\mathcal{X}}$, then [7]

$$\dot{q}(\bar{t}) = \lim_{h \rightarrow 0} h^{-1}(q(\bar{t} + h) - q(\bar{t})) \in TP(\mathcal{X}). \quad (1.4)$$

Conversely, given a time $\bar{t} \in \mathcal{R}$, a probability function $\bar{p} \in \mathcal{P}(\mathcal{X})$ and a tangent vector $v \in TP(\mathcal{X})$, the curve

$$t \mapsto \bar{p} + (t - \bar{t})v \in \mathcal{A}(\mathcal{X}) \quad (1.5)$$

has derivative v at $t = \bar{t}$.

1.1.3 Acceleration of the Curve

The *acceleration* of the curve $q(\cdot)$ is defined by

$$t \mapsto \ddot{q}(t) \in T\mathcal{P}(\mathcal{X}) . \quad (1.6)$$

If t acceleration is null we say that the curve is a *geodesic*. A locally length-minimizing curve is a geodesic. Equivalently, it is a path that a particle that is not accelerating would follow [5].

In this case, it happens if, and only if, the curve is a *mixture*, that is, $q, r \in \mathcal{A}(\mathcal{X})$ and

$$t \mapsto (1 - t)q + tr \in \mathcal{A}(\mathcal{X}) . \quad (1.7)$$

Given a function $\Phi: \mathcal{A}(\mathcal{X}) \rightarrow \mathcal{R}$, the *gradient* $\text{grad } \Phi$ is a mapping from $\mathcal{A}(\mathcal{X})$ to $T\mathcal{P}(\mathcal{X})$ such that for all smooth curve $q(\cdot)$, it holds

$$\frac{d}{dt}\Phi(q(t)) = \langle \text{grad } \Phi(q(t)), \dot{q}(t) \rangle \quad (1.8)$$

Notice that the *Grad* depends on the definition of velocity and on the definition of the scalar product. In this case, the componentwise computation is, given a numbering of $\mathcal{X} = [1 : n]$,

$$\begin{aligned} \frac{d}{dt}\Phi(q(1;t), \dots, q(n;t)) = \\ \sum_{j=1}^n \frac{\partial}{\partial q(j)} \Phi(q(1;t), \dots, q(n;t)) \dot{q}(j;t) = \\ \sum_{j=1}^n \left(\frac{\partial}{\partial q(j)} \Phi(\dots q(j;t) \dots) - \frac{1}{n} \sum_{k=1}^n \frac{\partial}{\partial q(k)} \Phi(\dots q(k;t) \dots) \right) \dot{q}(j;t) \end{aligned} \quad (1.9)$$

That is, $\text{grad} \neq \nabla$!

The previous construction provides a geometry of the probability simplex. However, *such a geometry is not the geometry used in Fisherian statistics*. You can see Part I of the monograph [9] for a nice presentation of the different styles of classical statistical inference, frequentist, Bayesian, and Fisherian. In turn, Fisher's methods are based on the methodology originally developed in *Genetics* and *Statistical Mechanics*. A reference to this last topic is, for example, the first part of [3] Hence, we start again with a different affine geometry. We will use elementary notions of differential geometry. Here is a one-slide summary of the definition of a differentiable manifold. Given a set M and a real vector space V , a *chart* is a 1-to-1 mapping.

$$s: U \rightarrow V, \text{ where } U \subset M \text{ and } s(U) \text{ is open in } V. \quad (1.10)$$

Below are the two charts s_1 and s_2 in the equation 1.11

$$s_1: U_1 \rightarrow V_1 \quad \text{and} \quad s_2: U_2 \rightarrow V_2 \quad (1.11)$$

These two charts are C^k -compatible if the change-of-chart mapping

$$s_2 \circ s_1^{-1}: s_1(U_1 \cap U_2) \rightarrow s_2(U_1 \cap U_2) \quad (1.12)$$

is a C^k diffeomorphism of open sets. A family of C^k -compatible charts is a C^k -atlas. A maximal class of equivalent atlases is a *manifold*. Here we have used the non-parametric version of differential geometry as presented in [2, 5].

1. **Manifold:** Consider a set M which has topology "Defines what are open sets and allows one to make sense of continuous maps, i.e map is continuous if the pre-image open set is open"
2. **Coordinate Chart:** Let $\phi: U \subset M \rightarrow \text{open set of } \mathbb{R}^d$, which is often denote by (U, ϕ) . Often we drop the U and just to refer ϕ for (U, ϕ) . We can see the below diagram.

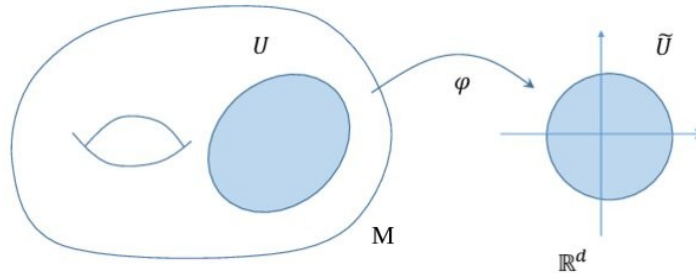


Figure 1.2: Coordinate chart ϕ applied on a smooth manifold M from an open set U to $\tilde{U}$ [14]

In the above figure 1.3 we have only one single chart. Now we want to cover the entire manifold with coordinate patches so that the union of the domains of the definition of the coordinate charts covered the entire manifold.

The coordinate chart allows us to define some notion of differentiability in terms of the coordinate representation of s_1 which is $s_2 \circ s_1^{-1}$, but we want to be independent of the choice of chart.

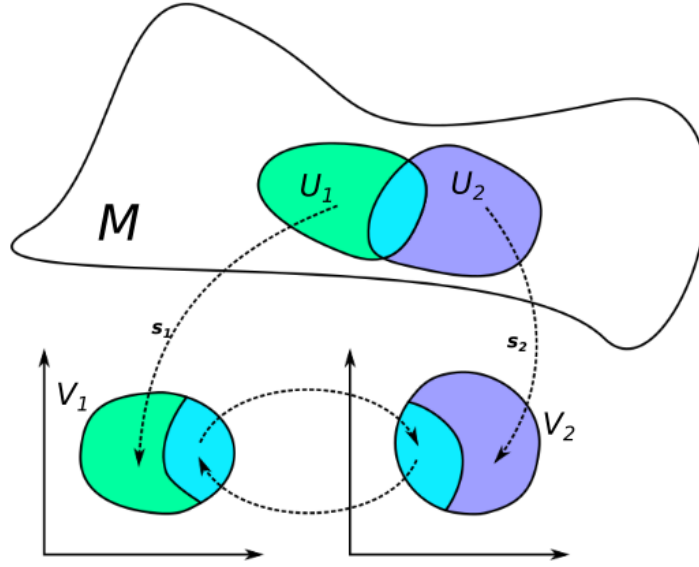


Figure 1.3: Change of Coordinate maps: Two charts are C^k -compatible if the change-of-chart mapping.

3. C^∞ **Atlas on Manifold M** is collection of charts (U_α, ϕ_α) of M such that:

- $\bigcup_\alpha U_\alpha = M$.
- For any pair of α and β with $U_\alpha \cap U_\beta \neq \emptyset$, then the $\phi_\alpha(U_\alpha \cap U_\beta)$ and $\phi_\beta(U_\alpha \cap U_\beta)$ are open in $\mathbb{R}^d$ and the change of the coordinate maps $\phi_\beta \circ \phi_\alpha^{-1}: \phi_\alpha(U_\alpha \cap U_\beta) \rightarrow \phi_\beta(U_\alpha \cap U_\beta)$ is C^∞ atlas. Now we can say that the elements of an atlas overlap smoothly.

4. Two atlas are equivalent if the $A_1 \cup A_2$ again an atlas. Given an atlas A , Let A^+ bet the set of all charts (U, ϕ) such that $A \cup \{(U, \phi)\}$ is also an atlas. It is to check that A^+ is also an atlas and we called the maximal atlas or complete atlas which is generated by A . Two atlas are equivalent if they generate the same maximal atlas. So, the maximal atlas an M is a differentiable structure on M .

1.1.4 Affine Spaces

Given a set M and a real finite dimensional vector space V , an *affine space* $([1])(M, V, \vec{\cdot})$ consists of a *displacement* mapping

$$M \times M \ni (p, q) \mapsto \overrightarrow{pq} \in V,$$

such that

- for each p the mapping $s_p: M \ni q \mapsto s_p(q) = \vec{pq}$ is 1-to-1, and
- the *parallelogram law*, $\vec{pq} + \vec{qr} = \vec{pr}$, that is, $s_p(q) + s_q(r) = s_p(r)$, holds.

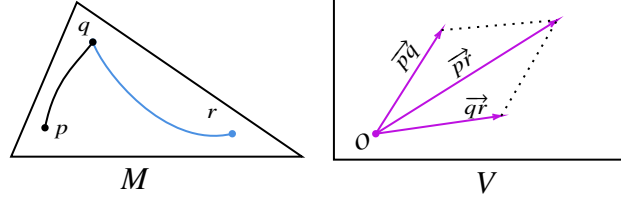


Figure 1.4: Affine space

1.1.5 Affine manifold

In particular, $\vec{pp} = s_p(p) = 0$ and $\vec{pq} + \vec{qp} = s_p(q) + s_q(p) = 0$. s_p is a *chart* if the image is open. The *atlas of charts* $s_p: M \rightarrow V$, $p \in M$, defines an *affine manifold*. An affine manifold is a manifold, but the atlas defining the affine manifold is made up of special functions [13] All change-of-charts mappings of the atlas are vector translations associated with a change of origin:

$$s_p \circ s_q^{-1}: v \mapsto s_p(q) + v \quad (1.13)$$

If $t \mapsto \rho(t) \in M$ is a curve, for each origin p the curve $t \mapsto s_p(\rho(t))$ is V -valued. The derivative $\frac{d}{dt}s_p(\rho(t))$ does not depend on p because

$$\frac{d}{dt}s_p(\rho(t)) = \frac{d}{dt}(s_p(q) + s_q(\rho(t))) = \frac{d}{dt}s_q(\rho(t)) \quad (1.14)$$

$\frac{D}{dt}\rho(t) = \frac{d}{dt}s_p(\rho(t))$ is, by definition, the *affine velocity*.

Example 1: Let us consider the base set M which is probability simplex $\mathcal{P}(\mathcal{X})$ and the vector space V is the tangent space of the simplex,

$$V = T\mathcal{P}(\mathcal{X}) = \left\{ v \in \mathbb{R}^{\mathcal{X}} \left| \sum_{x \in \mathcal{X}} v(x) = 0 \right. \right\}. \quad (1.15)$$

The displacement is $\vec{pq} = q - p \in V$ because $\sum_x (q(x) - p(x)) = 1 - 1 = 0$. The mapping $s_p(q) = q - p$ is 1-to-1. The parallelogram law is $(q - p) + (r - q) = (r - p)$. A *vector base*

of V produces a faithful *parametrization* of M . For example, a base of $TP_{>}(\mathcal{X})\{1, 2, 3\}$ is $v_1 = (1, -1, 0)$, $v_2 = (1, 0, -1)$, and

$$\begin{aligned} (p, q) \mapsto s_p(q) = -(q(2) - p(2)) \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - (q(3) - p(3)) \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \mapsto \\ (- (q(2) - p(2)), - (q(3) - p(3))) \in \\ \text{hull}((1, 0), (1, -1), (0, -1), (-1, 0), (-1, 1), (0, 1)) \subset \mathbb{R}^2 \end{aligned} \quad (1.16)$$

Example 2: Open Probability Simplex In the previous example, the image of each $s_p: q \mapsto q - p$ is a translation by $-p$ of the simplex. Hence it is closed in the tangent space, and the manifold structure is not well defined. We need an open image. Let us take as the base set the open simplex

$$\mathcal{P}_{>}(\mathcal{X}) = \{q \in \mathcal{P}(\mathcal{X}) \mid q(x) > 0, x \in \mathcal{X}\} \quad (1.17)$$

and the same vector space as in the previous case,

$$V = TP(\mathcal{X}) = TP_{>}(\mathcal{X}) \quad (1.18)$$

The displacement $s_p(q) = q - p$ satisfies the parallelogram law, and the image of s_p is $\mathcal{P}_{>}(\mathcal{X}) - p$ hence it is open in the tangent space. Hence, s_p is a chart. The inverse is $s_p^{-1}(v) = v + p$ and the change-of-chart is

$$v \mapsto s_p^{-1}(v) = v + p \mapsto s_q \circ s_p^{-1}(v) = (v + p) - q = v + s_q(p) \quad (1.19)$$

In the practice of Statistical Physics and Fisher's Statistics, other affine geometries imposed themselves[5].

1.1.6 Fisher Information

Let $X_1, X_2, \dots, X_n$ are random samples. Mathematically,

$$I(\theta) = E \left\{ \frac{\partial^2}{\partial \theta^2} \ell(X|\theta) \right\} \quad (1.20)$$

Fisher Score

Fisher Score in information geometry is defined by Aitchison's geometry by using a richer atlas of charts. There is a relation between Fisher's arguments and Statistical Physics.

A *curve* or *one-dimensional model* is a mapping

$$I \ni \theta \mapsto q(\theta) \in \mathcal{P}(\mathcal{X}) \quad (1.21)$$

As $\mathcal{P}(\mathcal{X}) \subset \mathbb{R}^{\mathcal{X}}$, we can assume I open and the mapping differentiable. In such a case,

$$\dot{q}(t) = \lim_{h \rightarrow 0} h^{-1}(q(t+h) - q(t)) \in T\mathcal{P}(\mathcal{X}) . \quad (1.22)$$

Precisely because of that, we use the term “tangent space”: the definition of tangent space depends on the definition of velocity of variation of the curve.

Let us suppose that there exists a couple $(x, \theta_0) \in \mathcal{X} \times I$ such that the curve $\theta \mapsto q(\theta)$ hits the facet $\{p \mid p(x) = 0\}$ at $\theta = \theta_0$, that is $q(x; \theta_0) = 0$. The differentiable curve $\theta \mapsto q(x, \theta)$ has a minimum at θ_0 , hence $\dot{q}(x, \theta_0) = 0$.

It holds $q(x; \theta) = 0 \Rightarrow \dot{q}(x; \theta) = 0$, hence and exists a curve $t \mapsto \dot{q}^*(t) \in \mathbb{R}^{\mathcal{X}}$, the *Fisher's score*, such that

$$\dot{q}(x; \theta) = \dot{q}^*(x; \theta) q(x; \theta) . \quad (1.23)$$

It is an *absolute continuity* setting, $\dot{q}(\theta) \ll q(\theta)$ [10]. In particular, if $\theta \mapsto q(\theta) \in \mathcal{P}_{>}\mathcal{X}$, then the *Fisher's score* is $\dot{q}^*(\theta) = \frac{d}{d\theta} \log q(\theta)$. There are many statistical reasons to take the Fisher score as a useful notion of the velocity of variation in the parameters of a statistical model [9, 4].

1.2 Statistical Bundle, Contrast Bundle

Let us consider for each $q \in \mathcal{P}_{>}\mathcal{X}$ the vector space of *q-contrasts*, that is, the random variables which are centered for q ,

$$B_q = \left\{ u \in \mathbb{R}^{\mathcal{X}} \left| \sum_{x \in \mathcal{X}} u(x) q(x) = 0 \right. \right\} . \quad (1.24)$$

The *statistical bundle* of $\mathcal{P}_{>}\mathcal{X}$ is the bundle $S\mathcal{P}_{>}\mathcal{X} \rightarrow \mathcal{P}_{>}\mathcal{X}$ of spaces of contrasts,

$$S\mathcal{P}_{>}\mathcal{X} = \{(q, v) \in \mathcal{P}_{>}\mathcal{X} \times \mathbb{R}^{\mathcal{X}} \mid \mathbb{E}_q[u] = 0\} \quad (1.25)$$

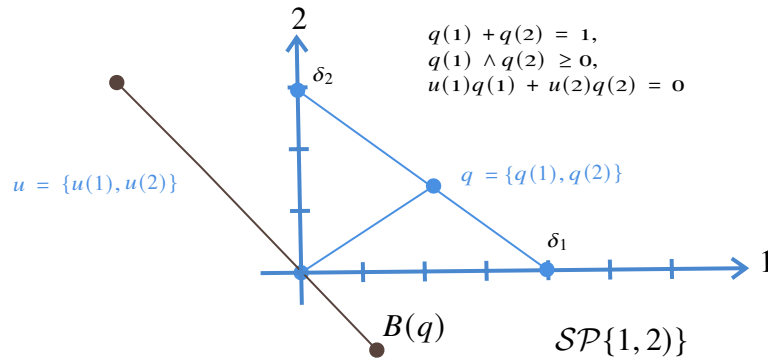


Figure 1.5: Statistical Bundle

1.3 Parallel Transport

In differential geometry, affine connection ∇ defines a way to “translate” vectors = “parallel transport” vectors on tangent planes along a smooth curve $c(t)$. (Manifolds are not embedded) Parallel transport of vectors between tangent planes T_p and T_q linked by c (defined by ∇), with $c(0) = p$ and $c(1) = q$:

$$\forall v \in T_p = \prod_{c(0) \mapsto c(1)}^{\nabla} v \in T_{c(t)} \quad (1.26)$$

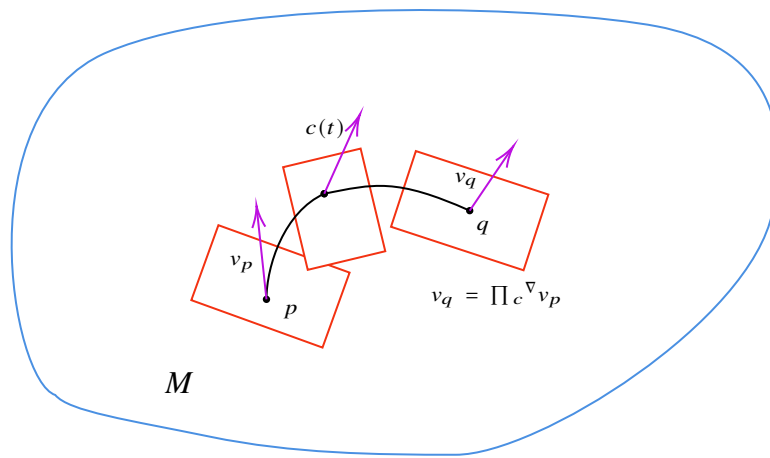


Figure 1.6: Parallel Transport

Each fiber B_q represents the tangent vectors (velocities) at q . We need to introduce "connections" between different "expressions" of the tangent space. Each fiber B_q has an inner product $\langle u, v \rangle_q = \mathbb{E}_q [uv]$. The mapping

$$S^2\mathcal{P}_>(\mathcal{X}) \ni (q, u, v) \mapsto \langle u, v \rangle_q \quad (1.27)$$

is called the *metric* of the statistical bundle.

Definition: For all $(p, u), (q, v) \in S\mathcal{P}_>\mathcal{X}$,

1. ${}^e\mathbb{U}_p^q: B_p \ni u \mapsto u - \mathbb{E}_q[u] \in B_q$ is the *exponential transport*;
2. ${}^m\mathbb{U}_q^p: B_q \ni v \mapsto \frac{q}{p}v \in B_p$ is the *mixture transport*.

Duality: $\forall (p, u), (q, v) \in S\mathcal{P}_>\mathcal{X}$,

$$\langle u, {}^m\mathbb{U}_q^p v \rangle_p = \langle {}^e\mathbb{U}_p^q u, v \rangle_q \quad (1.28)$$

Proof:

$$\begin{aligned} \langle u, {}^m\mathbb{U}_q^p v \rangle_p &= \left\langle p, \frac{q}{p}v \right\rangle_p = \mathbb{E}_p \left[uv \frac{q}{p} \right] = \\ &= \mathbb{E}_q [uv] = \mathbb{E}_q [(u - \mathbb{E}_q[u])v] = \langle u - \mathbb{E}_q[u], v \rangle_q = \langle {}^e\mathbb{U}_p^q u, v \rangle_q \end{aligned} \quad (1.29)$$

1.3.1 Affine Spaces with Parallel Transport

Because of what was said about the notion of Fisher's score and the bundle of contrasts, it is useful to rephrase Weyl's definition by moving from the trivial tangent bundle $M \times V$ to the nontrivial bundle of contrasts or fluctuations.¹

Definition: Affine Space Let M be a set and $B_p, p \in M$, a family of *topological vector spaces* (top-linear spaces). Let $({}^0\mathbb{U}_p^q), p, q \in M$ be a *cocycle* of top-linear isomorphism ${}^0\mathbb{U}_p^q: B_p \rightarrow B_q, {}^0\mathbb{U}_q^p {}^0\mathbb{U}_p^q = I_{B_p}$. Define a *displacement* mapping

$$\mathbb{S}: M \times M \ni (p, q) \mapsto s_p(q) \in B_p \quad (1.30)$$

so that:

1. For each fixed p the mapping $s_p: q \mapsto s_p(q) = \mathbb{S}(s, p)$ is *injective*
2. (Parallelogram law) $\mathbb{S}(p, q) + {}^0\mathbb{U}_q^p \mathbb{S}(q, r) = \mathbb{S}(p, r)$

¹While still staying in the finite state space case, I start using a more general language taken from infinite-dimensional differential geometry. See [5]

1.4 Affine manifold

The affine space provides an atlas of charts $s_p: M \rightarrow B_p$, $p \in M$. The *change-of-chart* map is $s_p \circ s_q^{-1}$. At $\rho = s_q^{-1}(w)$, $w \in B_q$, it holds

$$s_p \circ s_q^{-1}(w) = s_p(\rho) = s_p(q) + {}^0\mathbb{U}_q^p s_q(\rho) = s_p(q) + {}^0\mathbb{U}_q^p w. \quad (1.31)$$

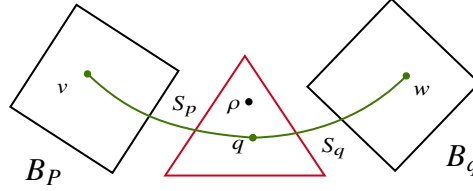


Figure 1.7: Affine Manifold

The change-of-origin map is the restriction of an affine map whose linear part is the parallel transport.

Assume that the vector fibers of the affine space $(M, (B_p p), \mathbb{S})$ are Banach spaces and assume that for each p , $s_p M$ is a neighborhood of 0 in B_μ . Define $U_p = s_p^{-1}(s_p(M)^\circ)$. Then $(s_p: U_p)$ is a chart on M . The charts are compatible, and the resulting manifold is the *affine manifold* of the affine space.

1.5 Affine Bundle

The specific form of the atlas defining the affine manifold allows the extension of the same atlas to define an affine bundle.

Given the affine manifold M , consider the set

$$\{(p, v) \mid p \in M, v \in B_q\} \quad (1.32)$$

and, for each $p \in M$ define the chart

$$s_p(q, v) = (s_p(q), {}^0\mathbb{U}_p^q v) \in B_p \times B_p \quad (1.33)$$

to define the manifold $S M$.

The statistical bundle of the affine manifold is where we define the velocity of a curve. The affine bundle has a tangent bundle where we define the second-order geometry that is, accelerations.

1.6 Exponential affine space

- We define the exponential displacement on $\mathcal{P}_>(\mathcal{X})$ by

$$\mathcal{P}_>(\mathcal{X}) \times \mathcal{P}_>(\mathcal{X}) \ni (p, q) \mapsto s_p(q) = \log \frac{q}{p} - \mathbb{E}_p \left[\log \frac{q}{p} \right] \in B_p, \quad (1.34)$$

and the exponential transport by

$${}^e\mathbb{U}_p^q : B_p \ni v \mapsto v - \mathbb{E}_q[v] \in B_q. \quad (1.35)$$

- The (generalized) parallelogram law is

$$\begin{aligned} \left(\log \frac{q}{p} - \mathbb{E}_p \left[\log \frac{q}{p} \right] \right) + {}^e\mathbb{U}_q^p \left(\log \frac{r}{q} - \mathbb{E}_q \left[\log \frac{r}{q} \right] \right) = \\ \left(\log \frac{q}{p} - \mathbb{E}_p \left[\log \frac{q}{p} \right] \right) + \left(\log \frac{r}{q} - \mathbb{E}_p \left[\log \frac{r}{q} \right] \right) = \\ \log \frac{r}{p} - \mathbb{E}_p \left[\log \frac{r}{p} \right]. \end{aligned} \quad (1.36)$$

- The inverse chart is defined on all of B_p by

$$s_p^{-1}(v) = \exp\{v - K_p(v)\} \cdot p = e_p(v) \quad K_p(v) = \log \mathbb{E}_p[e^v]. \quad (1.37)$$

- The *cumulant functional* K_p has several important well-known properties.

$$1. \quad D(p \| q) = \mathbb{E}_p \left[\log \frac{p}{q} \right] = \mathbb{E}_p \left[\log \frac{p}{\exp\{v - K_p(v)\} \cdot p} \right] = K_p(v), \text{ that is,}$$

$$K_p(s_p(q)) = D(p \| q). \quad (1.38)$$

$$2. \quad dK_p(v)[h] = \mathbb{E}_{e_p(v)}[h] = \left\langle \frac{e_p(v)}{p} - 1, h \right\rangle_{e_p(v)}$$

$$3. \quad d^2K_p(v)[h, k] = \text{Cov}_{e_p(v)}(h, k) = \left\langle {}^e\mathbb{U}_p^{e_p(v)} h, {}^e\mathbb{U}_p^{e_p(v)} k \right\rangle_{e_p(v)} = \left\langle h, \left({}^m\mathbb{U}_{e_p(v)}^p {}^e\mathbb{U}_p^{e_p(v)} \right) k \right\rangle_p.$$

1.7 Mixture affine space

- We define the *mixture displacement* on $\mathcal{P}_{>}(\mathcal{X})$ by

$$\mathcal{P}_{>}(\mathcal{X}) \times \mathcal{P}_{>}(\mathcal{X}) \ni (p, q) \mapsto \eta_p(q) = \frac{q}{p} - 1 \in B_p ,$$

and the *mixture transport* by

$${}^m\mathbb{U}_p^q : B_p \ni v \mapsto \frac{p}{q} v \in B_q . \quad (1.39)$$

- The (generalized) parallelogram law is

$$\left(\frac{q}{p} - 1 \right) + \frac{q}{p} \left(\frac{r}{q} - 1 \right) = \left(\frac{r}{p} - 1 \right) .$$

- The inverse chart is defined for all $v > 1$, $w \in B_p$ by

$$\eta_p^{-1}(v) = (1 + v) \cdot p . \quad (1.40)$$

1.8 Aitchison geometry

If the base set is the positive probability simplex $\mathcal{P}_{>}(\mathcal{X})$ with tangent space $T\mathcal{P}_{>}(\mathcal{X})$, define the displacement mapping $s_p : \mathcal{P}_{>}(\mathcal{X}) \rightarrow T\mathcal{P}_{>}(\mathcal{X})$ by [12]

$$s_p(q) = \log \frac{q}{p} - \sum_{y \in X} \frac{1}{\#X} \log \frac{q(y)}{p(y)} = \log \frac{q}{p} - \mathbb{E} \left(\log \frac{q}{p} \right) . \quad (1.41)$$

The difference with the exponential charts is the centering that does not depend on p . The mapping s_p is surjective, with inverse

$$s_p^{-1}(u) = \exp(u - K(u)) \cdot p$$

$$K(u) = \log \sum_{y \in X} e^{u(y)} p(y) = \log \mathbb{E}_p [e^u] . \quad (1.42)$$

Notice the relationship between the two forms of the normalizing constant:

$$K(u) = \mathbb{E} \left(\log \frac{p}{q} \right) . \quad (1.43)$$

The parallelogram law holds:

$$\left(\log \frac{q}{p} - \mathbb{E} \left(\log \frac{q}{p} \right) \right) + \left(\log \frac{r}{q} - \mathbb{E} \left(\log \frac{r}{q} \right) \right) = \log \frac{r}{p} - \mathbb{E} \left(\log \frac{r}{p} \right) \quad (1.44)$$

If p is uniform, that is, $p(x) = D^{-1}$, $D = \#X$, then

$$s_{D^{-1}}(q)(x) = \log q(x) - \sum_{y \in X} \log q(y) = \log \frac{q(x)}{(\prod_{y \in X} q(y))^{1/\#X}} \quad (1.45)$$

Notice that this special chart is equal to a chart in the exponential atlas. The exponential affine and Aitchison geometry are equal, just presented in different atlases. Precisely, the expression of the tangent space is the statistical bundle in one case and the tangent space of the simplex in the other case.

In Aitchison's geometry, the coordinates given by the chart above are called *centered log-ratio coefficients* clr [8].

Any n -sequence (n -sample) $q_1, \dots, q_n \in \mathcal{P}_{>}(X)$ is mapped to an n -sequence in the space of coordinates (contrasts),

$$u_1 = \text{clr}(q_1), \dots, u_n = \text{clr}(q_n) \in T\mathcal{P}_{>}(X) . \quad (1.46)$$

The space of contrasts $T\mathcal{P}_{>}(X)$ is a vector space with an inner product $\langle u, v \rangle = \mathbb{E}(uv) = \text{Cov}(u, v)$ so that we can perform all those computations that require a vector space structure.

The sample mean is

$$\begin{aligned} \frac{1}{n} \sum_{j=1}^n u_j &= \frac{1}{n} \sum_{j=1}^n \text{clr}(q_j) = \\ &= \frac{1}{n} \sum_{j=1}^n \left(\log q_j - \sum_{y \in X} \log q_j(y) \right) = \\ &= \log \left(\prod_{j=1}^n q_j \right)^{1/n} - \sum_{y \in X} \log \left(\prod_{j=1}^n q_j(y) \right)^{1/n} = \\ &= \text{clr} \left(\frac{\left(\prod_{j=1}^n q_j \right)^{1/n}}{\mathbb{E} \left(\left(\prod_{j=1}^n q_j \right)^{1/n} \right)} \right) \quad (1.47) \end{aligned}$$

The sample variance is based on the squared norm

$$\begin{aligned} \langle \text{clr}(q), \text{clr}(p) \rangle &= \mathbb{E}(\text{clr}(q) \text{clr}(r)) = \\ &\mathbb{E}(\log q \log r) - \mathbb{E}(\log q) \mathbb{E}(\log r) = \text{Cov}(\log q, \log r) . \end{aligned} \quad (1.48)$$

The velocity is

$$\frac{D}{dt}q(t) = \frac{d}{dt} \text{clr}(q(t)) = \frac{\dot{q}(t)}{q(t)} - \mathbb{E}\left(\frac{\dot{q}(t)}{q(t)}\right) \quad (1.49)$$

The velocity curve $t \mapsto \frac{D}{dt}q(t)$ is a curve in a vector space. Its derivative is the *acceleration*

$$\frac{d}{dt} \frac{D}{dt}q(t) = \frac{\ddot{q}(t)}{q(t)} - \left(\frac{\dot{q}(t)}{q(t)}\right)^2 - \mathbb{E}\left(\frac{\ddot{q}(t)}{q(t)} - \left(\frac{\dot{q}(t)}{q(t)}\right)^2\right) \quad (1.50)$$

Notice that the velocity is a deviation from a mean value in the uniform probability. The corresponding intensity is the variance of $\frac{d}{dt} \log q(t)$ in the uniform probability. *This choice does not fit Fisher's method, see below.*

The curves of zero acceleration are the curves of constant velocity. We call such curves *geodesics*. A geodesic from $p = \text{clr}(u)$ and initial velocity v has the form

$$q(t) = \text{clr}^{-1}(tv + u) = \exp\{tu - K_p(u)\} \cdot p . \quad (1.51)$$

Given a definition of the inner product in the vector space of an affine space, the *gradient* of a real function Φ on the base set (scalar field) is a mapping $\text{grad } \Phi$ from the base set to the vector space, the tangent space, that is, a vector field, such the for all smooth curve q it holds

$$\frac{d}{dt}\Phi(q(t)) = \left\langle \text{grad } \Phi(q(t)), \frac{D}{dt}q(t) \right\rangle . \quad (1.52)$$

In Aitchison geometry, if $\text{clr}(q(t)) = u(t)$ and $q(t) = \text{clr}^{-1}(u(t))$, then

$$\begin{aligned} \frac{d}{dt}\Phi \circ \text{clr}^{-1}(u(t)) &= \mathbb{E}\left(\text{grad } \Phi \circ \text{clr}^{-1}(u(t))\dot{u}(t)\right) = \\ &\frac{1}{\#X} \sum_{x \in X} \text{grad } \Phi \circ \text{clr}^{-1}(u(x;t))\dot{u}(x;t) \end{aligned} \quad (1.53)$$

It follows that

$$\frac{\partial}{\partial u(x)}\Phi \circ \text{clr}^{-1}(u) = \frac{1}{\#X} \text{grad } \Phi \circ \text{clr}^{-1}(u)(x) \quad (1.54)$$

or

$$\frac{1}{\#X} \text{grad } \Phi(q) = \nabla \left(\Phi \circ \text{clr}^{-1} \right) \circ \text{clr} \quad (1.55)$$

1.9 Exercises

Example 1: Gradient of the Entropy The entropy is defined for $q \in \mathcal{P}_{>}(\mathcal{X})$ by

$$\mathcal{H}(q) = \mathbb{E}_q[-\log q] = \sum_{x \in X} -\log q(x) q(x)$$

As $-\log$ is strictly convex, and $q(x) < 1$,

$$0 < -\log(\mathbb{E}_q[q]) < \mathcal{H}(q) .$$

As $s(x) = -x \log x$ is strictly concave,

$$\frac{1}{\#X} \mathcal{H}(q) = \sum_{x \in X} \frac{1}{\#X} s(q(x)) \leq s\left(\sum_{x \in X} \frac{1}{\#X} q(x)\right) = \frac{1}{\#X} \log \#X .$$

If m is the uniform probability function,

$$\mathcal{H}(m) = \sum_{x \in X} \frac{1}{\#X} \log\left(\frac{1}{\#X}\right) = \log \#X .$$

The entropy function is maximal at the uniform probability function.

By definition of *gradient* of a scalar field $\Phi: \mathcal{P}_{>}(\mathcal{X}) \rightarrow \mathbb{R}$, for each smooth curve $t \mapsto q(t)$,

$$\begin{aligned} \left\langle \text{grad } \mathcal{H}(q(t)), \frac{D}{dt} q(t) \right\rangle_{q(t)} &= \frac{d}{dt} \mathcal{H}(q(t)) \\ &= \frac{d}{dt} \sum_{x \in X} s(q(x; t)) \\ &= \sum_{x \in X} s'(q(x; t)) \frac{d}{dt} q(x; t) \\ &= \sum_{x \in X} (-1 - \log(q(x; t))) \frac{D}{dt} q(x; t) q(x; t) \\ &= \mathbb{E}_{q(t)} \left[-(1 + \log(q(t))) \frac{D}{dt} q(t) \right] \end{aligned}$$

The gradient must be centered so that

$$\text{grad } \mathcal{H}(q) = -(1 + \log(q)) - \mathbb{E}_q[-(1 + \log(q))] = \boxed{-\log(q) - \mathcal{H}(q)}$$

Notice that $\text{grad } \mathcal{H}(m) = 0$.

The *gradient flow equation* is

$$\boxed{\frac{D}{dt}q(t) = -\text{grad } \mathcal{H}(q(t))}$$

which is a *replicator equation*[10]

$$\frac{d}{dt}q(x; t) = q(x; t) \left(-\log(q(x; t)) - \sum_{y \in X} -\log(q(y; t)) q(y; t) \right).$$

If $q(0) = q_0$, then $\dot{q}^*(0) = \frac{D}{dt}q(t)|_{t=0} = -\log(q_0) - \mathcal{H}(q_0)$.

A computation shows that the solution of the gradient flow is a (non-canonical) exponential family of the form

$$q(t) = \exp\{\theta(t)\dot{q}^*(0) - K_{q_0}(\theta(t)\dot{q}^*(0))\} \cdot q_0.$$

with $\theta(0) = 0$

The first step is to compute the velocity. From

$$\log q(t) = \theta(t)\dot{q}^*(0) - K_{q_0}(\theta(t)\dot{q}^*(0)) + \log q_0$$

and the equation for the derivative of the cumulant function, the velocity is

$$\dot{q}^*(t) = \dot{\theta}(t)\dot{q}^*(0) - \mathbb{E}_{q(t)} [\dot{\theta}(t)\dot{q}^*(0)] = \dot{\theta}(t) (-\log q_0 + \mathbb{E}_{q(t)} [\log q_0])$$

The entropy is

$$\begin{aligned} \mathcal{H}(q(t)) &= \mathbb{E}_{q(t)} [-\log q(t)] = \\ &\mathbb{E}_{q(t)} [-\theta(t)\dot{q}^*(0) - \log q_0] + K_{q_0}(\theta(t)\dot{q}^*(0)) \end{aligned}$$

The gradient of the entropy is

$$\begin{aligned} -\log q(t) - \mathcal{H}(q(t)) &= \\ \theta(t)(-\dot{q}^*(0) + \mathbb{E}_{q(t)} [\dot{q}^*(0)]) - \log q_0 + \mathbb{E}_{q(t)} [\log q_0] &= \\ (\theta(t) - 1) (\log q_0 - \mathbb{E}_{q(t)} [\log q_0]) \end{aligned}$$

The gradient flow equation is satisfied by

$$\dot{\theta}(t) = \theta(t) - 1 \Rightarrow \theta(t) = e^t - 1$$

which in turn gives

$$q(t) \propto q_0^{e^t}.$$

It is also interesting to compute the limits $t \rightarrow \pm\infty$. Precisely, $\lim_{t \rightarrow -\infty} q(t) = m$, and the other case follows from

$$\left(\frac{q_0(x)}{\max_x q_0(x)} \right)^{e^t} \rightarrow \begin{cases} 1 & \text{if } q_0(x) = \max_x q_0(x) \\ 0 & \text{if } q_0(x) < \max_x q_0(x) \end{cases}$$

Fisher Information

$\theta \mapsto q(\theta)$ smooth curve $\dot{q}(\theta) = \frac{d}{d\theta} \log q(\theta)$ and $\mathbb{E}_{q(\theta)} [\dot{q}(\theta)] = 0$

Fisher Information:-

$$\mathcal{I}(\theta) = \text{var}\{\dot{q}(\theta)\} = \mathbb{E}_{q(\theta)} [\dot{q}(\theta)^2]$$

Gradient of Φ :- $\Phi : \mathcal{P}_+(\mathcal{X}) \mapsto \mathbb{R}$

$$\frac{d}{dt} \Phi(q(\theta)) = \langle \text{grad } \Phi(q(\theta)), \dot{q}(\theta) \rangle_{q(\theta)}$$

Cramer-Rao inequality:-

$$\begin{aligned} \left\{ \frac{d}{dt} \Phi(q(\theta)) \right\}^2 &= \{ \langle \text{grad } \Phi(q(\theta)), \dot{q}(\theta) \rangle_{q(\theta)} \}^2 \\ &\leq \mathbb{E}_{q(\theta)} [\text{grad } \Phi(q(\theta))]^2 \mathbb{E}_{q(\theta)} [\dot{q}(\theta)^2] \end{aligned}$$

$$\boxed{\mathbb{E}_{q(\theta)} [\text{grad } \Phi(q(\theta))]^2 \geq \left\{ \frac{d}{d\theta} \Phi(q(\theta)) \right\}^2 \mathcal{I}(\theta)^{-1}}$$

Example 2: Fisher Information $\Phi(q) = \mathbb{E}_q [t]$ and t is an unbiased estimate of $\Phi(q)$.

The error is given by

$$\text{grad } \Phi(q) = t - \mathbb{E}_q [t] = t - \Phi(q)$$

The error variance will be

$$\mathbb{E}_q [\{\text{grad } \Phi(q)\}^2] = \text{var}_q(t)$$

The Cramer-Rao inequality is

$$\text{var}_{q(\theta)}(t) \geq \left\{ \frac{d}{d\theta} \Phi(q(\theta)) \right\}^2 \mathcal{I}(\theta)^{-1}$$

Example 2.1 The entropy is

$$\Phi(q) = -\mathbb{E}_q [l \log(q)]$$

In the subject of statistical physics

$$\text{grad } \Phi(q) = -\log(q) - \{-\mathbb{E}_q[\log(q)]\}$$

The Cramer-Rao inequality is

$$\text{var}_{q(\theta)} \log(q(\theta)) \geq \left\{ \frac{d}{d(\theta)} \Phi(q(\theta)) \right\}^2 \mathcal{I}(\theta)^{-1}$$

Exercise: Replicator equation

Cf the literature with that keyword [10]

Let be given a real function $f \in C^1$ and consider the function of $\mathcal{P}_{>\mathcal{X}}$ defined by

$$H_f(q) = \mathbb{E}_q[f \circ q] = \sum_{x \in \mathcal{X}} q(x) f(q(x))$$

For example, the entropy is $\mathcal{H}(q) = H_{-\log}(q)$. There are other entropies of interest. The Kullback-Leibler divergence (from $p \in \mathcal{P}_{>\mathcal{X}}$) is

$$q \mapsto D(q \| p) = H_f(q), \quad f(q) = \log \frac{q}{p}.$$

The Tsallis entropy is defined for $\alpha \in \mathbb{R}, \alpha \neq 1$ by

$$S_\alpha(q) = \frac{1}{\alpha - 1} \left(1 - \sum_{x \in \mathcal{X}} q(x)^\alpha \right) = \mathbb{E}_q \left[\frac{1 - q^{\alpha-1}}{\alpha - 1} \right] = H_f(q)$$

with $f(q) = \frac{1-q^{\alpha-1}}{\alpha-1}$. Note $\lim_{\alpha \rightarrow 1} \frac{1-q^{\alpha-1}}{\alpha-1} = -\log q$.

Other entropies, such as Renyi entropy are of a more general form.

Let us compute the gradient of H_f . By definition of gradient of a scalar field $\Phi: \mathcal{X} \rightarrow \mathbb{R}$, for each smooth curve $t \mapsto q(t)$, the inner product $\langle \text{grad } H_f(q(t)), \frac{D}{dt} q(t) \rangle_{q(t)}$ equals

$$\begin{aligned} \frac{d}{dt} H_f(q(t)) &= \frac{d}{dt} \sum_{x \in \mathcal{X}} q(x; t) f(q(x; t)) \\ &= \sum_{x \in \mathcal{X}} \dot{q}(x; t) f(q(x; t)) + q(x; t) f'(q(x; t)) \dot{q}(x; t) \\ &= \sum_{x \in \mathcal{X}} (f(q(x; t)) + q(x; t) f'(q(x; t))) \frac{D}{dt} q(x; t) q(x; t) \\ &= \mathbb{E}_{q(t)} \left[(f(q(x; t)) + q(x; t) f'(q(x; t))) \frac{D}{dt} q(t) \right] \end{aligned}$$

Let us define $F(q) = f(q) + qf'(q)$. The gradient is

$$\text{grad } H_f(q) = F(q) - \mathbb{E}_q [F(q)]$$

and the gradient flow equation of H_f is the replicator equation

$$\frac{D}{dt} q(t) = F(q(t)) - \mathbb{E}_{q(t)} [F(q(t))] .$$

If the initial position is $q(0) = p$, then $\dot{q}^*(0) = F(p) - H_F(p)$. ²

1.10 Exercises: Deformed Logarithm

The mechanism of affine spaces can be generalized to deformed logarithms.

Let us consider that mapping $A : R_t \mapsto R_t$ is non-decreasing and let us define the $\log_A x = \int_1^x \frac{du}{A(u)}$ and is a concave function.

Example 1:-

Let $A(u) = u$ and $\log x = \int_1^x \frac{du}{u}$ This is the ordinary log.

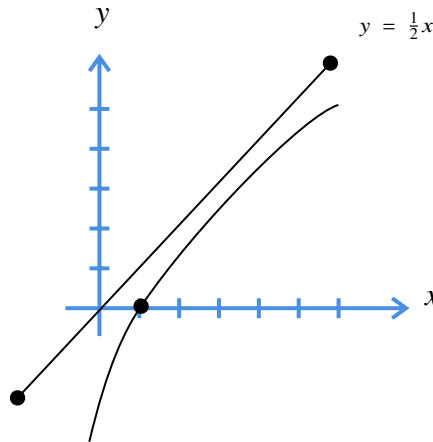
Example 2:-

Let $A(u) = 1$ and $\log_A x = \int_1^x \frac{du}{1} = x - 1$ This is the ordinary log.

Example 3:-

$$\frac{d}{dx} \left\{ \frac{1}{2} \right\} \left\{ x - \frac{1}{x} \right\} = \frac{1}{2} \left\{ 1 + \frac{1}{x^2} \right\}$$

$$A(u) = \frac{2u^2}{1+u^2} \text{ and } \log_A x = \int_1^x \frac{du}{A(u)}$$



²For example 6.2 of [10] and references therein.

Example 4:-

$$\log_A x = \frac{x^{1-\alpha} - 1}{1-\alpha}, \quad \alpha \neq 1, \quad \text{fnex } \alpha = \frac{1}{2}$$

$$\log_A x = 2\{x^{\frac{1}{2}} - 1\} = \int_1^x \frac{du}{\sqrt{u}}$$

Example 5:-

$$\log_A x = \frac{1}{2}(x + \log(x) - 1) = \int_1^x \frac{du}{\frac{2u}{1+u}}$$

1.10.1 Deformed ExponentialLet $\exp_A = \log_A^{-1}$ is convex function.**Example 1:-**

$$\log(x) = y \text{ and } y = \exp(x)$$

Example 2:-

$$\log_A(x) = x - 1 \text{ and } \exp_A(y) = 1 + y$$

Example 3:-

$$\frac{1}{2}\left\{x - \frac{1}{x}\right\} = y \iff x^2 - 2y^2 - 1 = 0 \text{ and } x > 0 \implies \exp_A(y) = y + \sqrt{1 + y^2}$$

Example 4:-

$$2(x^{\frac{1}{2}} - 1) = y, \quad \exp_A(y) = \left\{1 + \frac{1}{2}y^2\right\}$$

Example 5:-

$$\frac{1}{2}(x + \log(x) - 1) = y \iff 1 + 2y = \log(xe^x)$$

From the

$$\exp_A(\log_A(x)) = x \implies \exp'_A(y) = A(\exp_A(y))$$

1.10.2 Exponential Family(Non Parametric)

Let us consider that X is finite space, P is probability function, and u is random variable then

$$q = \exp_A(u - K_p(u) + \log_A(p))$$

is a probability function and $K_p(u)$ is the solution of

$$\sum_x \exp_A(u(x) - K_p(u) + \log_A p(x)) = 1$$

Now by computing the derivative $dK_p(u)[h] = \frac{d}{dt} K_p(u + th)|_{t=0}$

$$0 = \sum_x \exp_A(u(x) - K_p(u) + \log_A p(x)) \{h(x) - dK_p(u)[h]\}$$

$$= \sum_x q(x) \{h(x) - dK_p(u)[h]\} \text{ that is}$$

$$= \boxed{\mathbb{E}_{\tilde{q}}(h) = dK_p(u)[h]} \tilde{q} = \frac{A(q)}{\sum_x A(q(x))} \mathbb{E}_{\tilde{q}}(u) = 0$$

1.10.3 Affine Spaces

Define the bundle of the probability function

$$\{(q, v) | q \in \mathcal{T}(X), \mathbb{E}(\tilde{v}) = 0\} \text{ and the charts}$$

$$s_p(q) = \log_A(q) - \log_A(p) - \mathbb{E}_{\tilde{p}}(\log_A(q) - \log_A(p))$$

Example 1:-

$$s_p(q) = \frac{1}{2} \left\{ q - \frac{1}{q} \right\} - \frac{1}{2} \left\{ p - \frac{1}{p} \right\} - \sum_x \frac{2p(x)^2}{1 + 2p(x)^2} \left\{ \frac{1}{2} \left\{ q(x) - \frac{1}{q(x)} \right\} - \frac{1}{2} \left\{ p(x) - \frac{1}{p(x)} \right\} \right\} \times \frac{1}{\sum_x \frac{2p(x)^2}{1 + 2p(x)^2}}$$

1.10.4 Divergence

$$s_p^{-1}(u) = \exp_A(u - K_p(u) = \log_A(p)) \text{ and } u \in B_p$$

The deformed KL divergence will be

$$D(p \parallel q) = \mathbb{E}_{\tilde{q}}(\log_A(p) - \log_A(q))$$

Example 1:-

$$\sum_x \frac{2p(x)}{1+p(x)}(p(x) - q(x)) + \log \frac{p(x)}{q(x)}$$

1.10.5 Score

The velocity is

$$\begin{aligned} \frac{d}{dt} s_p(q(H)) &= \frac{d}{dt} \left\{ \log_A q(t) - \log_A(p) - \mathbb{E}_{\tilde{p}}(\log_A q(t) - \log_A(p)) \right\} \\ &= \frac{\dot{q}(t)}{A(q(t))} - \mathbb{E}_{\tilde{q}} \frac{\dot{q}(t)}{A(q(t))} \end{aligned}$$

The score is the velocity of the moving frame $p = q(t)$

$$\begin{aligned} \frac{D}{dt} \left\{ \frac{\dot{q}(t)}{A(q(t))} - \mathbb{E}_{\tilde{q}} \frac{\dot{q}(t)}{A(q(t))} \right\} &= \frac{\dot{q}(t)}{A(q(t))} \\ \mathbb{E}_{\tilde{q}} \frac{\dot{q}(t)}{A(q(t))} &= \sum \frac{q(x;t)}{A(q(x;t))} \frac{A(q(x;t))}{\sum_y A(q(y;t))} = 0 \end{aligned}$$

Chapter 2

Kinematics

2.1 Exponential Affine Space

Definition [Exponential Affine Space]

We define the exponential displacement on $\mathcal{P}_>(\mathcal{X})$ by

$$\mathcal{P}_>(\mathcal{X}) \times \mathcal{P}_>(\mathcal{X}) \ni (p, q) \mapsto s_p(q) = \log \frac{q}{p} - \mathbb{E}_1 \left[\log \frac{q}{p} \right] \in B_p,$$

and the exponential transport by

$${}^e\mathbb{U}_p^q : B_p \ni v \mapsto v - \mathbb{E}_q[v] \in B_q .$$

The (generalized) parallelogram law is

$$\begin{aligned} \left(\log \frac{q}{p} - \mathbb{E}_p \left[\log \frac{q}{p} \right] \right) + {}^e\mathbb{U}_q^p \left(\log \frac{r}{q} - \mathbb{E}_q \left[\log \frac{r}{q} \right] \right) = \\ \left(\log \frac{q}{p} - \mathbb{E}_p \left[\log \frac{q}{p} \right] \right) + \left(\log \frac{r}{q} - \mathbb{E}_p \left[\log \frac{r}{q} \right] \right) = \\ \log \frac{r}{p} - \mathbb{E}_p \left[\log \frac{r}{p} \right] . \end{aligned}$$

The inverse chart is defined on all of B_p by

$$s_p^{-1}(v) = \exp(v - K_p(v)) \cdot p = e_p(v) , \quad K_p(v) = \log \mathbb{E}_p[e^v] . \quad (2.1)$$

The *cumulant functional* K_p has several important properties. It is an expression of the KL divergence as a function of the second variable.

$$D(p \parallel q) = \mathbb{E}_p \left[\log \frac{p}{q} \right] = \mathbb{E}_p \left[\log \frac{p}{\exp((v - K_p(v)) \cdot p)} \right] = K_p(v), \text{ that is,}$$

$$K_p(v) = D(p \parallel e_p(v)) \quad K_p(s_p(q)) = D(p \parallel q) \quad (2.2)$$

The derivatives of K_p encode all of the features of the geometry.¹

2.1.1 Examples

Example 2.1.1 (First Derivative)

$$dK_p(v)[h] = \mathbb{E}_{e_p(v)}[h] = \left\langle \frac{e_p(v)}{p} - 1, h \right\rangle_p$$

$dK_p(v)[h]$ is the derivative of K_p at v in the direction h , that is,

$$\begin{aligned} dK_p(v)[h] &= \left. \frac{d}{dt} K_p(v + th) \right|_{t=0} = \left. \frac{d}{dt} \log \mathbb{E}_p [e^{v+th}] \right|_{t=0} \\ &= \mathbb{E}_p [e^{v+th}]^{-1} \mathbb{E}_p [h e^{v+th}] \Big|_{t=0} \\ &= \mathbb{E}_p [e^v]^{-1} \mathbb{E}_p [h e^v] = \mathbb{E}_{e_p(v)}[h] \\ &= \mathbb{E}_p \left[\frac{e_p(v)}{p} h \right] = \left\langle \frac{e_p(v)}{p} - 1, h \right\rangle_p \end{aligned}$$

Example 2.1.2 (Second Derivative)

$$d^2 K_p(v)[h, k] = \text{Cov}_{e_p(v)}(h, k) = \left\langle e \mathbb{U}_p^{e_p(v)} h, e \mathbb{U}_p^{e_p(v)} k \right\rangle_{e_p(v)} = \left\langle h, \left({}^m \mathbb{U}_{e_p(v)}^p e \mathbb{U}_p^{e_p(v)} \right) k \right\rangle_p.$$

$$\begin{aligned} d^2 K_p(v)[h, k] &= \left. \frac{d}{dt} dK_p(v + tk)[h] \right|_{t=0} \\ &= \left. \frac{d}{dt} \left\langle \frac{e_p(v + tk)}{p} - 1, h \right\rangle_p \right|_{t=0} \\ &= \left\langle \left. \frac{d}{dt} e^{v+tk - K_p(v+tk)} \right|_{t=0}, h \right\rangle_p \\ &= \left\langle \frac{e_p(v + th)}{p} \left(k - \frac{d}{dt} K_p(v + tk) \right) \Big|_{t=0}, h \right\rangle_p \\ &= \left\langle \frac{e_p(v)}{p} \left(k - \mathbb{E}_{e_p(v)}[k] \right), h \right\rangle_p = \text{Cov}_{e_p(v)}(h, k) \end{aligned}$$

¹Because of that, it is possible to derive the geometry from the cumulant function. This construction is called *Hessian geometry*.

Example 2.3 (Third Derivative)

$$d^3 K_p(v)[h_1, h_2, h_3] = \mathbb{E}_{e_p(v)} \left[\prod_{j=1}^3 \left(e \mathbb{U}_p^{e_p(v)} h_j \right) \right]$$

$$\begin{aligned}
d^3 K_p(v)[h_1, h_2, h_3] &= \frac{d}{dt} dK_p(v + th_3)[h_1, h_2] \Big|_{t=0} \\
&= \frac{d}{dt} \left\langle h_1, {}^m\mathbb{U}_{e_p(v+th_3)}^p e \mathbb{U}_p^{e_p(v+th_3)} h_2 \right\rangle_p \Big|_{t=0} \\
&= \left\langle h_1, \frac{d}{dt} \frac{e_p(v+th)}{p} (h_2 - \mathbb{E}_{e_p(v+th)} [h]_2) \Big|_{t=0} \right\rangle_p \\
&= \left\langle h_1, \frac{e_p(v)}{p} (h_3 - \mathbb{E}_{e_p(v)} [h]_3) (h_2 - \mathbb{E}_{e_p(v)} [h]_2) \right\rangle_p \\
&\quad - \left\langle h_1, \frac{e_p(v)}{p} \text{Cov}_{e_p(v)}[h_2, h_3] \right\rangle_p \\
&= \left\langle h_1, \mathbb{E}_{e_p(v)} [h]_1, (h_2, \mathbb{E}_{e_p(v)} [h]_2) (h_3, \mathbb{E}_{e_p(v)} [h]_3) \right. \\
&\quad \left. - \mathbb{E}_{e_p(v)} \{ (h_2, \mathbb{E}_{e_p(v)} [h]_2) (h_3, \mathbb{E}_{e_p(v)} [h]_3) \} \right\rangle_{e_p(v)} \\
&= \text{Cov}_{e_p(v)}[h_1] \{ (h_2, \mathbb{E}_{e_p(v)} [h]_2) (h_3, \mathbb{E}_{e_p(v)} [h]_3) \} \\
&= \mathbb{E}_{e_p(v)} \{ (h_1, \mathbb{E}_{e_p(v)} [h]_1) (h_2, \mathbb{E}_{e_p(v)} [h]_2) (h_3, \mathbb{E}_{e_p(v)} [h]_3) \}
\end{aligned}$$

2.2 Mixture affine space

We define the *mixture displacement* on $\mathcal{P}_>(\mathcal{X})$ by

$$\mathcal{P}_>(\mathcal{X}) \times \mathcal{P}_>(\mathcal{X}) \ni (p, q) \mapsto \eta_p(q) = \frac{q}{p} - 1 \in B_p ,$$

and the *mixture transport* by

$${}^m\mathbb{U}_p^q : B_p \ni v \mapsto \frac{p}{q} v \in B_q .$$

The (generalized) parallelogram law is

$$\left(\frac{q}{p} - 1\right) + \frac{q}{p} \left(\frac{r}{q} - 1\right) = \left(\frac{r}{p} - 1\right) .$$

The inverse chart is defined for all $v > 1$, $w \in B_p$ by

$$\eta_p^{-1}(v) = (1 + v) \cdot p . \quad (2.3)$$

2.3 Kinematics

The *velocity* of the smooth curve $t \mapsto \gamma(t)$ of the affine manifold is the curve $t \mapsto (\gamma(t), \dot{\gamma}^*(t))$ of the affine bundle whose second component is

$$\dot{\gamma}^*(t) = \lim_{h \rightarrow 0} h^{-1} (s_{\gamma(t)}(\gamma(t+h)) - s_{\gamma(t)}(\gamma(t))) = \left. \frac{d}{dh} s_{\gamma(t)}(\gamma(t+h)) \right|_{h=0} .$$

The expression of the velocity in the chart centred at p is

$$\left. \frac{d}{dt} s_p(\gamma(t)) \right|_{h=0} = \left. \frac{d}{dh} s_p(\gamma(t+h)) \right|_{h=0}$$

The velocity is the expression of the derivative in a moving reference centred at the curve itself. That is,

$${}^0\mathbb{U}_p^{\gamma(t)} \frac{d}{dt} s_p(\gamma(t)) = \dot{\gamma}^*(t) \text{ and } {}^0\mathbb{U}_{\gamma(t)}^p \dot{\gamma}^*(t) = \frac{d}{dt} s_p(\gamma(t)) .$$

The mapping

$$t \mapsto (\gamma(t), \dot{\gamma}^*(t)) \in SM$$

is called *lifting* of γ . If the origin is p , the expression of the lifting is

$$\left(s_p(\gamma(t)), {}^0\mathbb{U}_{\gamma(t)}^p \dot{\gamma}^*(t) \right) = \left(s_p(\gamma(t)), \frac{d}{dt} s_p(\gamma(t)) \right) =$$

Let F be a section of the affine bundle (vector field), that is, $(p, F(p)) \in SM$, $p \in M$. An *integral curve* of the section F is a curve $t \mapsto \gamma(t)$ such that $\dot{\gamma}^*(t) = F(\gamma(t))$. A *flow* of the section F is a mapping

$$M \times I \ni (v, t) \mapsto \Gamma_t(v)$$

such that for each v the curve $t \mapsto \Gamma_t(v)$ is an integral curve and $\gamma(0, v) = v$. A curve $I: t \mapsto \gamma(t)$ is *auto-parallel* in the affine bundle if

$$\dot{\gamma}^*(t) = {}^0\mathbb{U}_{\gamma(s)}^{\gamma(t)} \dot{\gamma}^*(s) \quad s, t \in I .$$

The following conditions are equivalent.

1. The curve γ is autoparallel.
2. The expression of the curve in each chart is affine.
3. For all s, t , $\gamma(t) = s_{\gamma(s)}^{-1}((t-s)\dot{\gamma}^*(s))$.

From $\dot{\gamma}^*(t) = {}^0\mathbb{U}_{\gamma(s)}^{\gamma(t)} \dot{\gamma}^*(s)$, apply ${}^0\mathbb{U}_{\gamma(t)}^p$, to get

$${}^0\mathbb{U}_{\gamma(t)}^p \dot{\gamma}^*(t) = {}^0\mathbb{U}_{\gamma(t)}^p {}^0\mathbb{U}_{\gamma(s)}^{\gamma(t)} \dot{\gamma}^*(s) = {}^0\mathbb{U}_{\gamma(s)}^p \dot{\gamma}^*(s)$$

that is, $\frac{d}{dt}s_p(\gamma(t)) = \frac{d}{ds}s_p(\gamma(s))$. The derivative is constant, hence

$$s_p(\gamma(t)) = s_p(\gamma(s)) + (t-s) \frac{d}{dt}s_p(\gamma(s)) .$$

Especially, $s_{\gamma(s)}(\gamma(t)) = (t-s)\dot{\gamma}^*(s)$.

Consider the curve $t \mapsto \gamma(t)$ with velocity $t \mapsto \dot{\gamma}^*(t)$. The *acceleration* $t \mapsto \ddot{\gamma}^*(t)$ is the velocity $t \mapsto (\mu(t), \dot{\gamma}^*(t))$. That is,

$$t \mapsto (\gamma(t), \dot{\gamma}^*(t), \ddot{\gamma}^*(t)) \in S^2M = \{(q, v, w) \mid v, w \in S_qM\}$$

and

$$\ddot{\gamma}^*(t) = \left. \frac{d}{dh} {}^0\mathbb{U}_{\gamma(t+h)}^{\gamma(t)} \dot{\gamma}^*(t+h) \right|_{h=0} \quad (2.4)$$

A curve is autoparallel if, and only if, the acceleration is null.

2.3.1 Examples of Kinematics

Example 2.3.1 (Velocity in the mixture affine manifold)

$$\ddot{p}^*(t) = \lim_{h \rightarrow 0} h^{-1} \mathbb{S}(p(t), p(t+h)) = \lim_{h \rightarrow 0} h^{-1} \left(\frac{p(t+h)}{p(t)} - 1 \right) = \frac{\dot{p}(t)}{p(t)} \quad (2.5)$$

Example 2.3.2 (Velocity in the exponential affine manifold)

$$\begin{aligned} \ddot{p}^*(t) &= \lim_{h \rightarrow 0} h^{-1} (p(t), p(t+h)) = \\ &= \lim_{h \rightarrow 0} h^{-1} \left(\log \frac{p(t+h)}{p(t)} - \sum \log \frac{p(t+h)}{p(t)} p(t) \right) = \frac{\dot{p}(t)}{p(t)} \end{aligned}$$

It is remarkable that the expression of the velocity is the same in both cases. However, auto-parallel curves are not the same. In the mixture case

$$\gamma(t) = s_{\gamma(s)}^{-1}((t-s)\dot{\gamma}^*(s)) = (1 + (t-s)\dot{\gamma}^*(s)) \cdot \gamma(s) = (\gamma(s) + (t-s)\dot{\gamma}(s)) . \quad (2.6)$$

In the exponential case

$$\gamma(t) = s_{\gamma(s)}^{-1}((t-s)\dot{\gamma}^*(s)) = \exp((t-s)\dot{\gamma}^*(s) - K_p((t-s)\dot{\gamma}^*(s))) \cdot \gamma(s) . \quad (2.7)$$

The exponential velocity for a curve of the form of a Gibbs model $p(t) \propto e^{\alpha(t)u} \cdot p$ that is $p(t) = e^{\alpha(t)u - \psi(t)} \cdot p$, is

$$\dot{p}^*(t) = \frac{d}{dt}(\alpha(t)u - \psi(t)) = \dot{\alpha}(t)u - \dot{\psi}(t) = \dot{\alpha}(t) \left(u - \sum u p(t) \right) . \quad (2.8)$$

In this case, the quantity $\dot{p}^*(t)$ is seen as $\dot{\alpha}(t)$ times the *fluctuation* $\left(U - \int U p(t) dm \right)$.

$$p(t)^{**} = m\mathbb{U}_p^{p(t)} \frac{d}{dt} m\mathbb{U}_{p(t)}^p \dot{p}^*(t) = \frac{p}{p(t)} \frac{d}{dt} \frac{p(t)}{p} \frac{\dot{p}(t)}{p(t)} = \frac{\ddot{p}(t)}{p(t)}$$

In the exponential case, the acceleration is

$$\begin{aligned} p(t)^{**} &= e\mathbb{U}_p^{p(t)} \frac{d}{dt} e\mathbb{U}_{p(t)}^p \dot{p}^*(t) = e\mathbb{U}_p^{p(t)} \frac{d}{dt} \left(\frac{\dot{p}(t)}{p(t)} - \int \frac{\dot{p}(t)}{p(t)} p dm \right) = \\ &= \frac{\ddot{p}(t)}{p(t)} - \left(\frac{\dot{p}(t)}{p(t)} \right)^2 - \int \left(\frac{\ddot{p}(t)}{p(t)} - \left(\frac{\dot{p}(t)}{p(t)} \right)^2 \right) p(t) dm = \\ &= \frac{\ddot{p}(t)}{p(t)} - \left[\left(\frac{\dot{p}(t)}{p(t)} \right)^2 - \int \left(\frac{\dot{p}(t)}{p(t)} \right)^2 p(t) dm \right] \end{aligned}$$

For the Gibbs model above, the exponential acceleration is proportional to the velocity, namely

$$p(t)^{**} = \ddot{\alpha}(t) \left(u - \sum u p(t) \right) = \frac{\ddot{\alpha}(t)}{\dot{\alpha}(t)} \dot{p}^*(t) . \quad (2.9)$$

Notice the cases $\alpha(t) = t$, $\alpha(t) = \frac{1}{2}t^2$, $\alpha(t) = 1 + e^t$.

The auto-parallel curves (geodesics) in the *mixture* geometry are of the form

$$\gamma(t) = \gamma(0) + \dot{\gamma}(0)t = (1 + \dot{\gamma}^*(0))\gamma(0) = (1-t)\gamma(0) + t\gamma(1)$$

The last expression explains the name.

In the *exponential* geometry, the form of the auto-parallel curve (geodesic) is

$$\gamma(t) = S_{\gamma(0)}^{-1}(t(\dot{\gamma}(0))) = e^{t\dot{\gamma}(0) - K_{\gamma(0)}(\dot{\gamma}(0))} \cdot \gamma(0)$$

that is, it is an *exponential family* [9].

The auto-parallel (geodesic) interval is the *exponential arc*

$$\gamma(t) \propto \gamma(0)^{1-t} \gamma(1)^t$$

Notice that the exponential arc is open; the parameter t belongs to an open interval containing $[0, 1]$.

Chapter 3

Linear Programming

3.1 Linear Programming

The *primal problem in canonical form* is

$$\begin{aligned} \text{Find } c &= \inf \sum_x c(x)r(x) \\ \text{Subject to } \sum_x A(y,x)r(x) &= \beta(y) \quad y \in Y \\ r(x) &\geq 0 \end{aligned}$$

r is the *primal plan* a plan is *feasible* if the constraints hold. The *dual problem in standard form* is

$$\begin{aligned} \text{Find } \beta &= \sup \sum_y \beta(y)\lambda(y) \\ \text{Subject to } \sum_y A(y,x)\lambda(y) &\leq c(x) \end{aligned}$$

λ is the *dual plan*

Strong Duality Theorem

If a feasible primal plan exists, then $c = \beta$. If moreover, $c > -\infty$, then primal optimal and dual optimal plans exist.

3.1.1 Fixed Margins

Assume a product sample space $X = X_1 \times X_2$ and consider the probability simplex $\mathcal{P}(\mathcal{X}_1 \times \mathcal{X}_2)$. The two marginalization mappings are

$$\begin{aligned}\Pi_1: \mathcal{P}(\mathcal{X}_1 \times \mathcal{X}_2) \ni q(\cdot, \cdot) &\mapsto \sum_{x_2} q(\cdot, x_2) \in \mathcal{P}(\mathcal{X}_1) \\ \Pi_2: (\mathcal{X}_1 \times \mathcal{X}_2) \ni q(\cdot, \cdot) &\mapsto \sum_{x_1} q(x_1, \cdot) \in \mathcal{P}(\mathcal{X}_2)\end{aligned}$$

For each given $q_1 \in \mathcal{P}(\mathcal{X}_1)$ and $q_2 \in \mathcal{P}(\mathcal{X}_2)$ define

$$\Pi(q_1, q_2) = \{q \in \mathcal{P}(\mathcal{X}_1 \times \mathcal{X}_2) \mid \Pi_1 q = q_1, \Pi_2 q = q_2\} .$$

The set of *transport plans* $\Pi(q_1, q_2)$ is

- non-empty,
- convex,
- compact.

The marginalization conditions

$$\begin{aligned}\sum_{x_2 \in X_2} q(y_1, x_2) &= q_1(y_1) \quad y_1 \in X_1 \\ \sum_{x_1 \in X_1} q(x_1, y_2) &= q_2(y_2) \quad y_2 \in X_2\end{aligned}$$

can be written as

$$\begin{aligned}\sum_{(x_1, x_2) \in X_1 \times X_2} (y_1 = x_1) q(x_1, x_2) &= q_1(y_1) \\ \sum_{(x_1, x_2) \in X_1 \times X_2} (y_2 = x_2) q(x_1, x_2) &= q_2(y_2)\end{aligned}$$

thus identifying an operator $A: (X_1 \cup X_2) \times (X_1 \times X_2)$ with

$$\sum_{(x_1, x_2) \in X_1 \times X_2} A(y; x_1, x_2) q(x_1, x_2) = (q_1 \cup q_2)(y)$$

3.2 Kantorovich Optimal Transport

Given the *cost* $c: X_1 \times X_2 \rightarrow \mathbb{R}$, Kantorovich looks for the transport plan with minimal expected cost

$$\inf \left\{ \sum_{x_1, x_2} c(x_1, x_2) q(x_1, x_2) \mid q \in \Pi(q_1, q_2) \right\}$$

It is a primal problem in canonical form:

$$\begin{aligned} \text{Find } c &= \inf \sum_{x_1, x_2} c(x_1, x_2) q(x_1, x_2) \\ \text{Subject to } \sum_x A(y; x_1, x_2) q(x_1, x_2) &= (q_1 \cup q_2)(y) \\ q(x_1, x_2) &\geq 0. \end{aligned}$$

The dual problem in standard form is

$$\begin{aligned} \text{Find } \beta &= \sup \sum_y (q_1 \cup q_2)(y) \lambda(y) \\ \text{Subject to } \sum_y A(y; x_1, x_2) \lambda(y) &\leq c(x_1, x_2) \end{aligned}$$

that is, given the form of A ,

$$\begin{aligned} \text{Find } \beta &= \sup \left(\sum_{y_1 \in X_1} q_1(y_1) \lambda_1(y_1) + \sum_{y_2 \in X_2} q_2(y_2) \lambda_2(y_2) \right) \\ \text{Subject to } \lambda_1(x_1) + \lambda_2(x_2) &\leq c(x_1, x_2). \end{aligned}$$

There exists a feasible transport plan, and the set of plans is compact. It follows that the full strong duality theorem holds. Let $\bar{q}$, $\bar{\lambda}$ be the optimal plan and dual plan. The equality of values gives

$$\begin{aligned} \sum_{x_1, x_2} c(x_1, x_2) \bar{q}(x_1, x_2) &= \\ \sum_{x_1 \in X_1} q_1(x_1) \bar{\lambda}_1(x_1) + \sum_{x_2 \in X_2} q_2(x_2) \bar{\lambda}_2(x_2) &= \\ \sum_{x_1, x_2} (\bar{\lambda}_1(x_1) + \bar{\lambda}_2(x_2)) \bar{q}(x_1, x_2) \end{aligned}$$

Now, the inequality $c \geq \lambda_1 \oplus \lambda_2$ implies

$$c(x_1, x_2) = \bar{\lambda}_1(x_1) + \bar{\lambda}_2(x_2) \quad \text{provided } \bar{q}(x_1, x_2) \neq 0$$

3.3 Transport plans in Probability Simplex

Define for all $q_1 \in \mathcal{P}(\mathcal{X}_1)$ and $q_2 \in \mathcal{P}(\mathcal{X}_2)$

$$\overset{\circ}{\Pi}(q_1, q_2) = \{q \in \mathcal{P}(\mathcal{X}_1 \times (\mathcal{X}_2)) \mid \Pi_1 q = q_1, \Pi_2 q = q_2\}$$

This is a submanifold of the affine exponential manifold.

A *submanifold of the affine manifold* $(M, s_p, B_p, {}^0\mathbb{U}_p^q : p, q \in M)$ is a subset $N \subset M$ such that for each $q \in N$ there exists a splitting of the fibre $B_q = T_q N \oplus R_q N$ and a splitting of the image of the restriction of q to N that provide an atlas on N .

The space $S_q N$ is the affine expression of the tangent space to N . We consider an important example. Finite dimensional exponential families and finite dimensional mixture models are notable examples. Notice that the orthogonal projection on the tangent space of the model provides the required splitting. The transport model will provide a second example.

Tangent vectors to the set of plans If $t \mapsto q(t)$ is a smooth curve of $\mathcal{P}(\mathcal{X})$ with values in the set of strictly positive transport plans, $q(t) \in \overset{\circ}{\Pi}(q_1, q_2)$. then for each random variable depending on the first variable only, $f = f_1 \circ \Pi_1$ it holds

$$0 = \frac{d}{dt} \mathbb{E}_{q_1} [f_1] = \frac{d}{dt} \mathbb{E}_{q(t)} [f_1 \circ \Pi_1] = \langle f_1 \circ \Pi_1 - \mathbb{E}_{q(t)} [f_1 \circ \Pi_1], \dot{q}^*(t) \rangle_{q(t)} = \mathbb{E}_{q(t)} [f_1 \circ \Pi_1 \dot{q}^*(t)] ,$$

Similarly on the other projection. f_1 being generic, It follows that

$$\boxed{\mathbb{E}_{q(t)} [\dot{q}^*(t) | \Pi_1] = 0 \quad \text{and} \quad \mathbb{E}_{q(t)} [\dot{q}^*(t) | \Pi_2] = 0}$$

3.4 ANOVA

1. Effects

The linear sub-spaces of $L^2(q)$ which, respectively, express the q -grand-mean, the two q -simple effects, and the q -interactions, are

$$\begin{aligned} B_0(q) &\sim \mathbb{R}, \\ B_1(q) &= \{f \circ \Pi_1 \mid f_1 \in L_0^2(q_1)\}, \\ B_2(q) &= \{f_2 \circ \Pi_2 \mid f_2 \in L_0^2(q_2)\}, \\ B_{12}(q) &= (B_0(q) + B_1(q) + B_2(q))^\perp, \end{aligned} \tag{3.1}$$

where the orthogonality is computed in the q probability function, that is in the inner product of $L^2(q)$, $\langle u, v \rangle_q = \mathbb{E}_q[uv]$.

Each element of the space $B_0(q) + B_1(q) + B_2(q)$ has the form $u = u_0 + f_1 \circ \Pi_1 + f_2 \circ \Pi_2$, where $u_0 = \mathbb{E}_\gamma[u]$ and f_1, f_2 are uniquely defined.

2. Spitting

For each $q \in \mathcal{P}(\mathcal{X})$ there exist a unique orthogonal splitting

$$L^2(q) = \mathbb{R} \oplus (B_1(q) + B_2(q)) \oplus B_{12}(q) .$$

Namely, each $u \in L^2(\gamma)$ can be written uniquely as

$$u = u_0 + (u_1 + u_2) + u_{12} ,$$

where $u_0 = \mathbb{E}_\gamma[u]$ and $(u_1 + u_2)$ is the q -orthogonal projection of $u - u_0$ unto $(B_1(q) + B_2(q))$. That is,

$$\mathbb{E}_q[u_{12}|\Pi_2] = 0 \quad \text{and} \quad \mathbb{E}_q[u_{12}|\Pi_1] = 0$$

Let us write the ANOVA decomposition of the statistical bundle as

$$S_q \mathcal{P}_>(\mathcal{X}) = (B_1(q) + B_2(q)) \oplus B_{12}(q) .$$

3.4.1 Fiber of the sub-manifold

1. Let $t \mapsto q(t) \in \overset{\circ}{\Pi}(q_1, q_2)$ be a smooth curve with $q(0) = \gamma$. Then the velocity at q is an interaction, $\dot{q}(0) \in B_{12}(q)$.
2. Given any interaction $v \in B_{12}(q)$, the curve $t \mapsto q(t) = (1 + tv)q$ stays in $\overset{\circ}{\Pi}(q_1, q_2)$ for t in a neighborhood of 0 and, moreover, $v = \dot{q}(0)$.

3.5 Statistical Bundle of the Transport Plans

The SBTP with margins q_1 and q_2 is the sub-statistical bundle

$$\overset{\circ}{S\Pi}(q_1, q_2) = \left\{ (q, v) \mid q \in \overset{\circ}{\Pi}(q_1, q_2), \mathbb{E}_q[v|\Pi_1] = \mathbb{E}_\gamma[v|\Pi_1] = 0 \right\} .$$

The transport ${}^m\mathbb{U}_q^r$ maps the fiber at q to the fiber at r . The sub-manifold of the transport model is *flat* in the mixture geometry and there is no simple expression of the exponential coordinate. The splitting of the statistical bundle suggests a *mixed parametrization* of $\mathcal{P}_>(\mathcal{X})$. This is a classical topic in the statistics of contingency tables.

3.6 Gradient flow of the OT problem

Let us discuss the *Optimal Transport OT* problem in the framework of the transport model bundle. Let $c : X_1 \times X_2 \rightarrow \mathbb{R}$ be a cost function and define the expected cost function

$$C : \mathcal{P}_>(\mathcal{X}) \ni q \mapsto \mathbb{E}_q [c]$$

1. Gradient

The function $q \mapsto C(q)$ restricted to the open transport model $\overset{\circ}{\Pi}(q_1, q_2)$ has (statistical) gradient in $S\overset{\circ}{\Pi}(q_1, q_2)$ given by

$$\nabla C : q \mapsto c_{12,q} = c - c_{0,q} - (c_{1,q} + c_{2,q})$$

Gradient flow The equation of the *gradient flow of C* is

$$\dot{q}^* = - (c - c_{0,q} - (c_{1,q} + c_{2,q})) = -c_{12,q}$$

The gradient mapping $\nabla C(q)$ is defined to be the orthogonal projection of the cost c onto the space of q -interactions $B_{12}(\gamma)$.

Assume $\gamma \mapsto c_{12,q}$ is defined for all q .

2. Stationary point

If $\hat{q}$ is a zero of the extended gradient map, $\nabla C(\hat{q}) = 0$, then it holds

$$c(x, y) = c_{0,q} + c_{1,q}(x) + c_{2,q}(y), \quad (x, y) \in \text{Supp}(\hat{q}) .$$

We expect any solution $t \mapsto q(t)$ of the gradient flow equation to converge to a plan $\bar{q} = \lim_{t \rightarrow \infty} q(t) \in \Pi(q_1, q_2)$ such that $\mathbb{E}_{\bar{q}} [c]$ is the value of the Kantorovich optimal transport problem. The form of the gradient is compatible with the classical result in OT:

3.7 Exercise

Example 1

$$\delta \in \mathcal{P}(\mathcal{X}) X_{\circ} = \text{supp } f$$

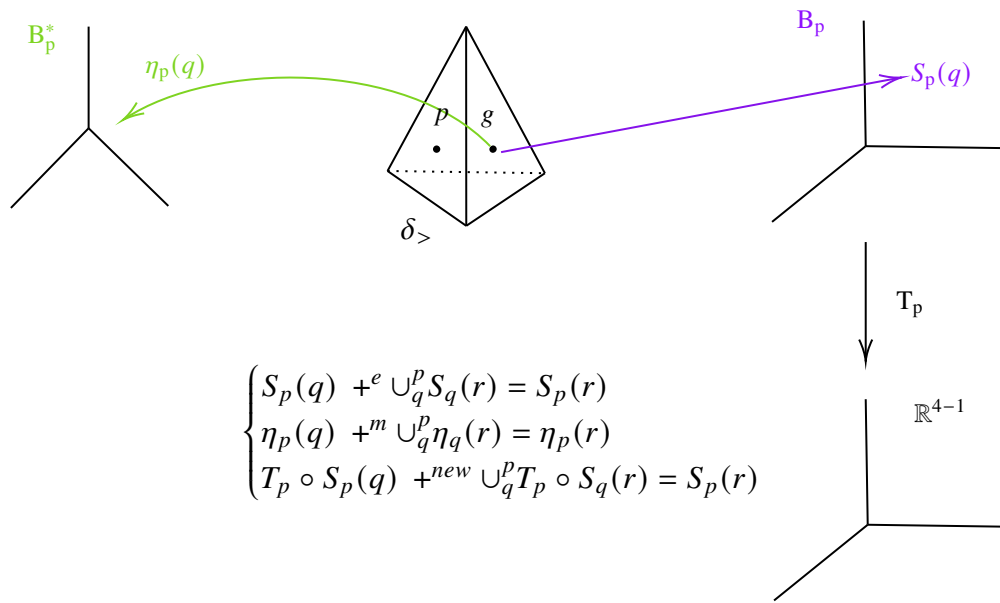


Figure 3.1: Affine Statistical Bundle

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